



INHOUD

WINDOWS COMMANDS	7
BASIC IP TROUBLESHOOTING COMMANDS	7
<i>Verifying IP-Address</i>	7
<i>Verification of End-To-End IPv4 Connectivity</i>	7
ADVANCED IP TROUBLESHOOTING COMMANDS	7
<i>Verifying DNS records</i>	7
<i>Resolve the domain name into IP address</i>	7
<i>Verifying ARP table ARP cache</i>	7
<i>Delete ARP cache</i>	7
<i>Troubleshoot of End-To-End IPv6 Connectivity</i>	7
REMOTE SESSIONS	7
<i>Remote TELNET connection in CMD</i>	7
<i>Remote SSH connection in CMD</i>	7
FACTORY DEFAULTS RESET	8
SWITCH WITH IOS RESET PROCEDURES	8
<i>Reset Catalyst IOS Switches to Factory Defaults in ROMMON mode</i>	8
<i>Reboot switch</i>	8
<i>Initial Configuration Dialog (if it is shown)</i>	8
SWITCH WITH IOS-XE RESET PROCEDURES	8
<i>Reset Catalyst IOS-XE Switches to Factory Defaults in ROMMON mode</i>	8
<i>Initial Configuration Dialog (if it is shown)</i>	8
<i>Erase config.text in privilege exec mode</i>	8
<i>Erase vlan.dat in privilege exec mode</i>	8
<i>Save empty configuration</i>	9
<i>Set IGNORE_STARTUP back to original value in ROMMON</i>	9
ROUTER RESET PROCEDURES	9
<i>Reset routers to Factory Defaults in ROMMON mode</i>	9
<i>Initial Configuration Dialog (if it is shown)</i>	9
<i>Save empty configuration</i>	9
RESET COMMANDS IN IOS PRIVILEGE EXEC MODE	9
<i>Erase config.text in privilege exec mode</i>	9
<i>Erase VLAN.dat in privilege exec mode</i>	9
<i>Restart the networking device</i>	9
<i>Initial Configuration Dialog</i>	9
NAVIGATE IOS	10



NAVIGATE IOS COMMANDS	10
<i>Navigate to / exit privilege exec mode</i>	10
<i>Enter global configuration mode</i>	10
<i>Go to lower mode</i>	10
BASIC COMMANDS	11
INITIAL DEVICE SETTINGS / COMMANDS	11
<i>Changing hostname</i>	11
<i>Configuring MOTD banner</i>	11
<i>Configuring password privilege exec mode</i>	11
<i>Encrypting passwords in configuration file</i>	11
<i>Securing console port</i>	11
<i>Securing terminal lines (Telnet) on a switch</i>	11
<i>Securing terminal lines (Telnet) on a router</i>	11
<i>Setting the default gateway on a switch</i>	11
<i>Saving the configuration to the NVRAM (startup-configuration → config.text)</i>	11
INTERFACE CONFIGURATION SWITCH	12
SELECTING AN INTERFACE ON A SWITCH	12
<i>Entering the SVI (Switch Virtual Interface) on a Switch for remote management</i>	12
<i>Entering a Layer 2 interface on a Switch</i>	12
CONFIGURING A IPV4-ADDRESS ON THE SVI	12
<i>Configure a IP- address</i>	12
DOCUMENTATE AN INTERFACE	12
<i>Placing a description on an interface</i>	12
<i>Enables an interface</i>	12
INTERFACE CONFIGURATION ROUTER	13
SELECTING AN INTERFACE ON A ROUTER	13
<i>Entering a Layer 3 interface on a Router</i>	13
<i>Entering a virtual interface on a Router</i>	13
CONFIGURING A IPV4-ADDRESS ON AN LAYER 3 INTERFACE, SVI OR LOOPBACK INTERFACE	13
<i>Configure a IP- address</i>	13
DOCUMENTATE AN INTERFACE	13
<i>Placing a description on an interface</i>	13
<i>Enables an interface</i>	13
OTHER INTERFACE CONFIGURATION	14
RESETTING AN INTERFACE TO DEFAULTS SETTINGS	14
<i>Resets an interface</i>	14



<i>Enables IPv4-routing on the MLS.....</i>	<i>14</i>
<i>Enables router port on the MLS.....</i>	<i>14</i>
IPv6 on aN INTERFACE	15
ENABLING IPV6 ON A SWITCH	15
<i>Enable IPv6 on a switch.....</i>	<i>15</i>
<i>Enables IPv6-routing on the router or the MLS.....</i>	<i>15</i>
CONFIGURING A GUA IPV6-ADDRESS ON AN LAYER 3 INTERFACE, SVI OR LOOPBACK INTERFACE	15
<i>Configure a GUA IPv6-address.....</i>	<i>15</i>
<i>Configure a IPv6 Stateless Autoconfiguration on a client</i>	<i>15</i>
<i>Configure a IPv6 Stateful Autoconfiguration on a client.....</i>	<i>15</i>
CONFIGURING A LINK-LOCAL IPV6-ADDRESS ON AN LAYER 3 INTERFACE, SVI OR LOOPBACK INTERFACE	15
<i>Configure a Link-local IPv6-address.....</i>	<i>15</i>
SECURE REMOTE ACCESS	16
SSH	16
<i>Generate RSA key pairs</i>	<i>16</i>
<i>Configure user authentication.....</i>	<i>16</i>
<i>Enable SSH version 2.....</i>	<i>16</i>
<i>Enable SSH on VTY lines.....</i>	<i>16</i>
MAINTAINING IOS	17
UPDATING IOS	17
<i>Upgrade Cisco IOS Image with TFTP.....</i>	<i>17</i>
DEVICE HARDENING	17
Network Device Security COMMANDS.....	17
<i>Inactivity timer.....</i>	<i>17</i>
<i>Limiting Remote Access with ACLs</i>	<i>17</i>
<i>Mitigating Brute-force attacks</i>	<i>17</i>
<i>Cisco AutoSecure</i>	<i>17</i>
<i>Translate a word to an IP address (DNS).....</i>	<i>17</i>
<i>Creating a VLAN</i>	<i>18</i>
<i>Assigning a Port to a VLAN.....</i>	<i>18</i>
<i>Assigning a Port to a Data VLAN</i>	<i>18</i>
<i>Disable the autonegotiation (Dynamic Trunking Protocol)</i>	<i>18</i>
<i>Configuring an 802.1Q Trunk</i>	<i>18</i>
<i>VLAN Trunk Protocol (VTP) Commands.....</i>	<i>19</i>
<i>Router on a Stick.....</i>	<i>19</i>



<i>Configuring EtherChannel LACP.....</i>	<i>19</i>
<i>Configuring a redundant router</i>	<i>19</i>
<i>Configure Single-Area OSPF.....</i>	<i>20</i>
<i>Configuring Standard IPv4 ACLs</i>	<i>20</i>
<i>Configuring Extended IPv4 ACLs</i>	<i>20</i>
<i>Configuring Static Translation of Inside Source Addresses.....</i>	<i>21</i>
<i>Configuring Dynamic Translation of Inside Source Addresses.....</i>	<i>21</i>
<i>Configuring IPv4 Inside PAT.....</i>	<i>21</i>
<i>Connects the interface to the inside or outside network.....</i>	<i>21</i>
<i>Switch VLAN Configuration to Support WLANs.....</i>	<i>21</i>
<i>Switch Port Connected to Autonomous AP Configuration</i>	<i>21</i>
<i>Syslog messages format</i>	<i>21</i>
<i>Syslog messages format</i>	<i>22</i>
<i>HTTP service.....</i>	<i>22</i>
<i>Internet Control Message Protocol.....</i>	<i>22</i>
<i>Verifying the integrity of the downloaded IOS image</i>	<i>22</i>
<i>Connecting a client with Telnet and SSH</i>	<i>22</i>
<i>Configuring a Router as an IPv4 DHCP Server</i>	<i>23</i>
<i>Configuring an IPv4 DHCP Relay agent</i>	<i>23</i>
<i>Configuring SLAAC on a router</i>	<i>23</i>
<i>Configuring IPv6 Stateless DHCPv6 on a router</i>	<i>23</i>
<i>Configuring IPv6 Stateful DHCPv6 on a router</i>	<i>23</i>
<i>Configuring an IPv6 DHCP Relay agent on a router-interface.....</i>	<i>24</i>
<i>Disable or enable CDP Globally</i>	<i>24</i>
<i>Disable or enable CDP on interface only.....</i>	<i>24</i>
<i>Disable LLDP Globally</i>	<i>24</i>
<i>Disable or enable CDP on interface only.....</i>	<i>24</i>
<i>Host Static Routes.....</i>	<i>24</i>
<i>Configuring IPv4 Static Routes</i>	<i>24</i>
<i>Configuring IPv6 Static Routes</i>	<i>24</i>
<i>Spanning Tree Priority Root Bridge Optimization</i>	<i>25</i>
<i>Rapid Spanning Tree Protocol</i>	<i>25</i>
<i>BPDU guard</i>	<i>25</i>
<i>Configuring Port Security.....</i>	<i>25</i>
<i>Mitigating VLAN Attacks</i>	<i>25</i>
<i>Mitigating DHCP-related attacks</i>	<i>25</i>



<i>Mitigate the chances of ARP spoofing.....</i>	<i>26</i>
<i>Mitigating STP Attacks</i>	<i>26</i>
<i>Mitigating Broadcast storms.....</i>	<i>26</i>
<i>Configure Software Clock</i>	<i>26</i>
<i>Configuring NTP.....</i>	<i>26</i>
<i>Locating Cisco IOS Image Files.....</i>	<i>27</i>
SHOW COMMANDS	28
<i>Verify Basic Configuration</i>	<i>28</i>
<i>Inspect Privilege Level.....</i>	<i>28</i>
<i>Inspect TCP/IP Applications</i>	<i>28</i>
<i>Verify configured DHCP address pools</i>	<i>28</i>
<i>Verify configured DHCPv6 address pools.....</i>	<i>28</i>
<i>Checking Interface Configuration and Status.....</i>	<i>28</i>
<i>View Cisco Discovery Protocol (CDP) information</i>	<i>28</i>
<i>Verify LLDP (Link Layer Discovery Protocol).....</i>	<i>28</i>
<i>View the ARP table</i>	<i>28</i>
<i>View Logging</i>	<i>29</i>
<i>View neighbor discovery table.....</i>	<i>29</i>
<i>Verifying IPv6 Static Route Configuration</i>	<i>29</i>
<i>Verifying VLANs</i>	<i>29</i>
<i>Verifying the VLAN configuration of an interface</i>	<i>29</i>
<i>Verifying VTP configuration.....</i>	<i>29</i>
<i>Verifying MAC addresses.....</i>	<i>29</i>
<i>Verifying IPv4 routing table.....</i>	<i>29</i>
<i>Verify Single-Area OSPF.....</i>	<i>29</i>
<i>Verify EtherChannel Configuration.....</i>	<i>29</i>
<i>Verify STP information.....</i>	<i>29</i>
<i>Verify HSRP information</i>	<i>29</i>
<i>Verify the configured access-list.....</i>	<i>29</i>
<i>Verify Network Address Translation</i>	<i>31</i>
<i>Display the state of logging (syslog).....</i>	<i>31</i>
<i>Verifying NTP Status</i>	<i>31</i>
<i>Verifying running Services</i>	<i>31</i>
<i>Verifying Port Security</i>	<i>31</i>
<i>Verifying DHCP Snooping.....</i>	<i>31</i>
<i>Verifying Dynamic ARP Inspection</i>	<i>31</i>



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WINDOWS COMMANDS

BASIC IP TROUBLESHOOTING COMMANDS

Verifying IP-Address

C:\ipconfig /all	Displays the complete TCP/IP configuration for all adapters
C:\ipconfig /release	Sends a DHCPRELEASE message to the DHCP server
C:\ipconfig /renew	Renews DHCP configuration for all adapters

Verification of End-To-End IPv4 Connectivity

C:\ping ip-address domain name	A successful ping to an IPv4 address means that the endpoints have basic IPv4 connectivity between them
C:\tracert ip-address domain name	The results of traceroute to an IPv4 address can help you determine how far along the path data can successfully reach

ADVANCED IP TROUBLESHOOTING COMMANDS

Verifying DNS records

C:\ipconfig /displaydns	Displays the contents of the host DNS cache
C:\ipconfig /flushdns	Deletes the host DNS cache

Resolve the domain name into IP address

C:\nslookup domain name	Example: nslookup www.google.com
-------------------------	----------------------------------

Verifying ARP table | ARP cache

C:\arp -a	Displays the current ARP table for all interfaces
C:\arp -a -N ip_address	Limit the output of the arp command to a single interface

Delete ARP cache

C:\arp -d	to clear ARP cache
-----------	--------------------

OR:

C:\netsh interface IP delete arpcache	to clear ARP cache
---------------------------------------	--------------------

Troubleshoot of End-To-End IPv6 Connectivity

C:\netsh interface ipv6 show neighbors	Lists all devices that are currently in the IPv6 neighbor discovery table cache
--	---

REMOTE SESSIONS

Remote TELNET connection in CMD

C:\telnet ip-address	Attempts a TELNET sessions
----------------------	----------------------------

Remote SSH connection in CMD

C:\ssh -l username ip-address	Attempts a SSH sessions
-------------------------------	-------------------------

OR:

C:\ssh username@ip-address	Attempts a SSH sessions
----------------------------	-------------------------



FACTORY DEFAULTS RESET

SWITCH WITH IOS RESET PROCEDURES

Reset Catalyst IOS Switches to Factory Defaults in ROMMON mode

Power off the switch and hold down the **Mode button**. While holding down the Mode button power on the switch. Hold down the Mode button until you see the output **switch:**

NOTE: some switches displays the message "The password-recovery mechanism is enabled". Press MODE button twice before you see the output switch:

switch:flash_init	Loading flash
Switch:delete flash:config.text	Removes config.text
Switch:delete flash:vlan.dat	Removes vlan.dat

!-- Make sure vlan.dat and config.text both are deleted --!

Reboot switch

Switch:reset

Initial Configuration Dialog (if it is shown)

Device is empty when showing "Would you like to enter the initial configuration dialog [yes/no]" after restart.

Always enter **no**

SWITCH WITH IOS-XE RESET PROCEDURES

Reset Catalyst IOS-XE Switches to Factory Defaults in ROMMON mode

Power off the switch and hold down the **Mode button**. While holding down the Mode button power on the switch. Hold down the Mode button until wait for SYST and ACTV LED to become amber before releasing MODE button

switch:flash_init	Loading flash
switch:SWITCH_IGNORE_STARTUP_CFG=1	Bypass STARTUP-CONFIG in NVRAM
switch:boot flash:packages.conf	Reloads the switch

Initial Configuration Dialog (if it is shown)

Device is empty when showing "Would you like to enter the initial configuration dialog [yes/no]" after restart.

Always enter **no**

Erase config.text in privilege exec mode

SW#erase startup-config	Removes only config.text
-------------------------	--------------------------

OR:

SW#write erase	Removes only config.text
----------------	--------------------------

OR:

SW#delete flash:config.text	Removes only config.text
-----------------------------	--------------------------

Erase vlan.dat in privilege exec mode

SW#delete vlan.dat	Removes vlan.dat
--------------------	------------------

!-- Make sure vlan.dat and config.text both are deleted --!



Save empty configuration

SW#copy running-config startup-config	Copies the new and empty running-config to the old startup-config
SW#reload	
Set IGNORE STARTUP back to original value	
Hold the MODE button again during bootup	
Switch#no system ignore startupconfig switch all	Instructs the switch to read/load the startup-config
OR:	
Hold the MODE button again during bootup to enter ROMMON	
switch:SWITCH_IGNORE_STARTUP_CFG=0	Instructs the switch to read/load the startup-config

ROUTER RESET PROCEDURES

Reset routers to Factory Defaults in ROMMON mode

Press during bootup ALT-B (Tera term) when the router is just powercycled	Sent a break sequence to the router
rommon 1>confreg 0x2142	Prompt to boot from Flash without loading the configuration
rommon 2>reset	The router reboots but ignores its saved configuration

Initial Configuration Dialog (if it is shown)

Device is empty when showing "Would you like to enter the initial configuration dialog [yes/no]" after restart.

Always enter no

Save empty configuration

RT#copy running-config startup-config	Copies the new and empty running-config to the old startup-config
RT(config)#config-register 0x2102	Sets the Configuration register to its original setting 0x2102. During booting the startup-config will be loaded

RESET COMMANDS IN IOS PRIVILEGE EXEC MODE

Erase config.text in privilege exec mode

DEVICE#dir flash:	Shows content of NVRAM
DEVICE#erase startup-config	Removes only config.text
OR:	
DEVICE#write erase	Removes only config.text
OR:	
DEVICE#delete flash:config.text	Removes only config.text

Erase VLAN.dat in privilege exec mode

DEVICE#delete flash:vlan.dat	Removes vlan.dat
------------------------------	------------------

!-- Make sure vlan.dat and config.text both are deleted --!

Restart the networking device

DEVICE#reload

Initial Configuration Dialog

Device is empty when showing "Would you like to enter the initial configuration dialog [yes/no]" after restart.

Always enter no



NAVIGATE IOS

NAVIGATE IOS COMMANDS

Navigate to | exit privilege exec mode

DEVICE>**enable**

DEVICE#**disable**

Enter global configuration mode

DEVICE#**configure terminal**

Go to lower mode

exit	Navigates to previous mode
-------------	----------------------------

end	Navigates direct to privilege exec mode --> Switch#
------------	--



BASIC COMMANDS

INITIAL DEVICE SETTINGS / COMMANDS

Changing hostname

DEVICE(config)#**hostname** *name*

Configuring MOTD banner

DEVICE(config)#**banner motd** #ATTENTION! WE WILL BE
HAVING SCHEDULED SYSTEM MAINTENANCE ON THIS DEVICE.##

Message of the day (MOTD) banner is displayed to all terminals that are connected and is useful for sending messages that affect all users (such as impending system shutdowns)

OTHER OPTIONAL BANNERS:

DEVICE(config)#**banner login** #ACCESS FOR AUTHORIZED
USERS ONLY. PLEASE ENTER YOUR USERNAME AND PASSWORD#

Defines a customized login banner to be displayed before the username and password login prompts

DEVICE(config)#**banner exec** #WELCOME TO ICTMBO#

Displayed after the user has logged in to the router. It is not displayed for reverse-telnet connections

Configuring password privilege exec mode

DEVICE(config)#**enable secret** *pass-value*

MD5 hash

OR:

DEVICE(config)#**enable algorithm-type sha256 secret**
pass-value

SHA-256 hash (minimum IOS 15.3)

Encrypting passwords in configuration file

DEVICE(config)#**service password-encryption**

Securing console port

DEVICE(config)#**line con** 0

DEVICE(config-line)#**password** *pass-value*

DEVICE(config-line)#**login**

Better to use **login local** in combination with a username

Securing terminal lines (Telnet) on a switch

SW(config)#**line vty** 0 15

SW(config-line)#**password** *pass-value*

SW(config-line)#**login**

A switch uses VTY 0 15 (total of 16 remote sessions)

Better to use **login local** in combination with a username

Securing terminal lines (Telnet) on a router

RT(config)#**line vty** 0 4

RT(config-line)#**password** *pass-value*

RT(config-line)#**login**

A router uses VTY 0 4 (total of 5 remote sessions)

Better to use **login local** in combination with a username

Setting the default gateway on a switch

SW(config)#**ip default-gateway** *ip-address router*

Saving the configuration to the NVRAM (startup-configuration → config.text)

DEVICE#**copy running-config startup-config**

Press **enter** to confirm filename

Destination filename [startup-config]?

Building configuration...

[OK]

OR:

DEVICE#**write**

Use **wr** as a short for write memory

Building configuration...

[OK]



INTERFACE CONFIGURATION SWITCH

SELECTING AN INTERFACE ON A SWITCH

Entering the SVI (Switch Virtual Interface) on a Switch for remote management

SW(config)# interface vlan <i>vlan-value</i>	Basic: VLAN1 Better: management VLAN number
--	--

Entering a Layer 2 interface on a Switch

SW(config)# interface type <i>interface-id</i>	To configure one interface
--	----------------------------

OR:

SW(config)# interface range type <i>interface-id</i> - <i>interface-id</i>	To configure multiple interfaces simultaneously
---	---

CONFIGURING A IPV4-ADDRESS ON THE SVI

Configure a IP- address

SW(config-if)# ip address <i>ip-address mask</i>	Type: Fastethernet Gigabitethernet Serial interface-id example: 0/1
--	--

OR:

SW(config-if)# ip address dhcp	Acquire an IPv4 address through DHCP
---	--------------------------------------

DOCUMENTATE AN INTERFACE

Placing a description on an interface

SW(config-if)# description <i>name-string</i>
--

ENABLING AN INTERFACE

Enables an interface

SW(config-if)# no shutdown	Interface goes from down to up
--	--------------------------------



INTERFACE CONFIGURATION ROUTER

SELECTING AN INTERFACE ON A ROUTER

Entering a Layer 3 interface on a Router

RT(config)# interface <i>type interface-id</i>	To configure one router-interface
---	-----------------------------------

OR:

RT(config)# interface range <i>type interface-id - type interface-id</i>	To configure multiple router-interfaces simultaneous
---	--

Entering a virtual interface on a Router

RT(config)# interface loopback <i>Loopback-value</i>	A virtual interface that resides on a router. Will never go "down"
---	--

CONFIGURING A IPV4-ADDRESS ON AN LAYER 3 INTERFACE, SVI OR LOOPBACK INTERFACE

Configure a IP- address

RT(config-if)# ip address <i>ip-address mask</i>	Type: Fastethernet Gigabitethernet Serial interface-id example: 0/1
---	--

OR:

RT(config-if)# ip address dhcp	Acquire an IPv4 address through DHCP
---------------------------------------	--------------------------------------

DOCUMENTATE AN INTERFACE

Placing a description on an interface

RT(config-if)# description <i>name-string</i>	
--	--

ENABLING AN INTERFACE

Enables an interface

RT(config-if)# no shutdown	Interface goes from down to up
-----------------------------------	--------------------------------



OTHER INTERFACE CONFIGURATION

RESETTING AN INTERFACE TO DEFAULTS SETTINGS

Resets an interface

DEVICE(config)# default interface type interface-id	Delete the configuration of an interface at ones
--	--

OR:

DEVICE(config)# default range interface type interface-id - interface-id	Delete the configuration of multiple interfaces at ones
---	---

ENABLE IP-ROUTING ON A MULTILAYER SWITCH (MLS)

Enables IPv4-routing on the MLS

MLS(config)# ip routing	Enable IPv4-routing on an MLS
--------------------------------	-------------------------------

Enables router port on the MLS

MLS(config-if)# no switchport	Sets a switchport to a router port
--------------------------------------	------------------------------------



IPV6 ON AN INTERFACE

ENABLING IPV6 ON A SWITCH

Enable IPv6 on a switch

SW(config)#**sdm prefer dual-ipv4-and-ipv6 default**

Enables the switch to operate in dual-stack environments by supporting IPv4 and IPv6

ENABLE IPV6-ROUTING ON A ROUTER AND MULTILAYER SWITCH (MLS)

Enables IPv6-routing on the router or the MLS

DEVICE(config)#**ipv6 unicast-routing**

Enables IPv6 routing

CONFIGURING A GUA IPV6-ADDRESS ON AN LAYER 3 INTERFACE, SVI OR LOOPBACK INTERFACE

Configure a GUA IPv6-address

DEVICE(config-if)#**ipv6 address ipv6-address/prefix-length**

Configures a global unicast address (GUA)

OR:

DEVICE(config-if)#**ipv6 address ipv6-prefix/prefix-length eui-64**

Configures an IPv6 address for an interface and enable IPv6 processing on the interface using an EUI-64 interface ID in the low order 64 bits of the address (host)

Example: **ipv6 address 2001:0db8:0:1::/64 eui-64**

OR:

DEVICE(config-if)#**ipv6 address autoconfig [default]**

Configures stateless autoconfiguration on the interface. If you add the default keyword, the router will install a default route

Configure a IPv6 Stateless Autoconfiguration on a client

DEVICE(config-if)#**ipv6 enable**

Creates an IPv6 link-local address on a interface

DEVICE(config-if)#**ipv6 address autoconfig**

Enables automatic configuration of IPv6 addresses using stateless autoconfiguration (SLAAC) on an interface and enables IPv6 processing on the interface

Configure a IPv6 Stateful Autoconfiguration on a client

DEVICE(config-if)#**ipv6 address dhcp**

To enable DHCP for an IPv6 client process and acquire an IPv6 address on an interface

CONFIGURING A LINK-LOCAL IPV6-ADDRESS ON AN LAYER 3 INTERFACE, SVI OR LOOPBACK INTERFACE

Configure a Link-local IPv6-address

DEVICE(config-if)#**ipv6 address fe80::1 link-local**

Configures a link-local address



SECURE REMOTE ACCESS

SSH

Generate RSA key pairs

DEVICE(config)#ip domain-name *name*

Configures a domain-name

DEVICE(config)#crypto key generate rsa general-keys
modulus *bits*

Generates a key to encrypt SSH traffic

Configure user authentication

DEVICE(config)#username *name* [*privilege level*]
secret *password*

Creates a local database username entry with MD5 hash

Privilege Level Security:

- **Level 0** - Read-only, Zero-level access only allows five commands- logout, enable, disable, help and exit
- **Level 1** - Read-only, User-level access allows you to enter in User Exec mode that provides very limited read-only access to the router and the ability to use "Ping" to test connectivity and "Reload" to restart the router
- **Level 15** - Full Access, Privilege level access allows you to enter in Privileged Exec mode and provides complete control over the router to all commands, such as the "Reload" command, and the ability to make configuration changes

NOTE: You can create the "middle ground" by defining arbitrary roles through customization of privilege levels 2 through 14. For this example, we'll enable privilege level 2, then reassign both "Ping" and "Reload" commands

Router(config)#enable secret level 2 0 cisco123! (*enabled Level 2 with password or cisco123!*)

Router(config)#privilege exec level 2 ping

Router(config)#privilege exec level 2 reload

After applying the changes, users must have a minimum privilege level of 2 or above to execute "ping" and "reload"

OR:

DEVICE(config)#username *name* [*privilege level*]
algorithm-type sha256 secret *password*

Creates a local database username entry with **SHA-256 hash** (minimum IOS 15.3)

Enable SSH version 2

DEVICE(config)#ip ssh version 2

Enables SSH version 2

Enable SSH on VTY lines

DEVICE(config-line)#transport input {ssh / telnet}

Specifies multiple input protocols

DEVICE(config-line)#login local

Enables local password checking at login time. Authentication is based on the username

DEVICE(config-line)#no password

Removes the unencrypted password used by Telnet



MAINTAINING IOS

UPDATING IOS

Upgrade Cisco IOS Image with TFTP

```
SW1:format flash:/  
SW1:copy tftp://{ip_address}/{file_name} flash:  
SW1:set BOOT flash://{filename}
```

DEVICE HARDENING

Network Device Security COMMANDS

Inactivity timer

DEVICE(config-line)# exec-timeout <i>minutes seconds</i>	An IOS device will disconnect a console or VTY user after 10 minutes of inactivity
---	--

Limiting Remote Access with ACLs

DEVICE(config-line)# access-class <i>access-list-number</i> in	Applies a ACL to the lines
---	----------------------------

Mitigating Brute-force attacks

DEVICE(config)# login { <i>on-failure</i> / <i>on-failure</i> } log	Generates syslogs on logins
---	-----------------------------

DEVICE(config)# security passwords min-length <i>length</i>	Sets the minimum number of characters for passwords
---	---

DEVICE(config)# login block-for <i>seconds</i> attempts <i>tries</i> within <i>seconds</i>	Configures the device for login parameters that help provide DoS detection
--	--

Cisco AutoSecure

RT(config)# auto secure	For Cisco routers, the Cisco AutoSecure feature can be used to assist securing the system
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Translate a word to an IP address (DNS)

DEVICE(config)# no ip domain-lookup	Disable the DNS translation process
--	-------------------------------------



VLAN COMMANDS

Creating a VLAN

SW1(config)# vlan <i>vlan-id</i>	
SW1(config-vlan)# name <i>vlan-name</i>	
SW1(config-vlan)# shutdown	Prevent traffic to flow on that VLAN. Used for preventing unused ports from communicating with each other

Assigning a Port to a VLAN

SW1(config-if)# switchport mode access	A static-access port can belong to one VLAN and is manually assigned to that VLAN
SW1(config-if)# switchport voice vlan <i>vlan-id</i>	A voice VLAN port is an access port attached to a Cisco IP Phone, configured to use one VLAN for voice traffic and another VLAN for data traffic from a device attached to the phone

Assigning a Port to a Data VLAN

SW1(config-if)# switchport access vlan <i>vlan_number</i>	Assigns the single VLAN to the interface. To reset the VLAN to the default VLAN, use the no form of this command
--	--

Disable the autonegotiation (Dynamic Trunking Protocol)

SW1(config-if)# switchport nonegotiate	Switches from other vendors do not support DTP. As discussed, DTP is used by Cisco switches to automatically negotiate whether an interface between two switches will be put into access or trunk mode. Best practice is to disable the autonegotiation and not use the dynamic auto and dynamic desirable switch port modes.
---	--

Configuring an 802.1Q Trunk

SW1(config)# switchport mode trunk	A trunk port is a member of all VLANs by default, including extended-range VLANs, but membership can be limited by configuring the allowed-VLAN list.
SW1(config-if)# switchport trunk native vlan <i>vlan_number</i>	Best practice is to disable the autonegotiation and not use the dynamic auto and dynamic desirable switch port modes
SW1(config-if)# switchport trunk native vlan tag	By default, the native VLAN is not tagged, but it is recommended to tag the native VLAN
SW1(config-if)# switchport trunk allowed vlan <i>vlan_list</i>	Sets allowed VLANs on a trunk link. <i>vlan_list</i> is a sequence of VLAN numbers, separated with commas. The <i>vlan_list</i> cannot have spaces after commas
SW1(config-if)# switchport trunk allowed vlan add <i>vlan_list</i>	Adds specified VLANs to the existing list of allowed VLANs on a trunk link. The <i>vlan_list</i> cannot have spaces after commas
SW1(config-if)# switchport trunk allowed vlan remove <i>vlan_list</i>	Removes specified VLANs from the existing list of allowed VLANs on a trunk link. The <i>vlan_list</i> cannot have spaces after commas.
SW1(config-if)# switchport trunk encapsulation dot1q	Tells the switch that the interface should use IEEE 802.1Q encapsulation on the frames when the interface is configured as a trunk



VLAN Trunk Protocol (VTP) Commands	
SW1(config)# vtp mode {client server transparent}	Configure VTP device mode
Server: In VTP server mode, you can create, modify, and delete VLANs and specify other configuration parameters (such as VTP version and VTP pruning) for the entire VTP domain Client: VTP clients behave the same way as VTP servers, but you cannot create, change, or delete VLANs on a VTP client. Transparent: VTP transparent network devices do not participate in VTP. However, in VTP version 2, transparent network devices do forward VTP advertisements that they receive out their trunking LAN ports	
SW1(config)# vtp domain domain_name	Set the name of the VTP administrative domain
SW1(config)# vtp password password_string	Set the password for the VTP administrative domain
SW1(config)# vtp version {1 2 3}	Set the administrative domain to VTP version

Inter-VLAN Routing COMMANDS	
Router on a Stick	
RT1(config)# interface subinterface	Enters subinterface configuration mode Example: interface G0/0/0.10
RT1(config-if)# encapsulation dot1q vlan_number [native]	Defines the encapsulation format as IEEE 802.1Q and specifies the VLAN identifier
RT1(config-if)# ip address ip_address network_mask	

EtherChannel Commands	
Configuring EtherChannel LACP	
MLS1(config)# interface port-channel port-channel-identifier	Creates the port channel logical interface
MLS1(config-if)# no switchport	By default, port channel interface is a Layer 2 interface. Therefore, use the no switchport command to make it a routed interface
MLS1(config-if)# channel-group channel-group-number mode {on} {active passive}	Assigns the interface to the port channel interface and automatically creates the port channel interface. The channel-group-number specifies the identifier of the port channel interface for the aggregated link

HSRP Commands	
Configuring a redundant router	
RT1(config-if)# standby [group-number] ip [ip-address] [secondary]	Activates the Hot Standby Router Protocol (HSRP)
RT1(config-if)# standby [group-number] priority priority	Priority value that prioritizes a potential Hot Standby router. The range is from 1 to 255, where 1 denotes the lowest priority and 255 denotes the highest priority
RT1(config-if)# standby [group-number] preempt [delay { [minimum seconds] [reload seconds] [sync seconds] }]	Controls preemption in the standby configuration for the initially active interface
RT1(config-if)# standby [group-number] track type interface-id	Allows you to specify another interface on the router for the HSRP process to monitor in order to alter the HSRP priority for a given group



OSPF COMMANDS

Configure Single-Area OSPF

RT1(config)# router ospf process-id	Enables OSPF routing and enters router configuration mode
RT1(config-router)# network ip-address wildcard-mask area area-id	Defines an interface on which OSPF runs and defines the area ID for that interface
RT1(config-router)# default-information originate [always]	Configures OSPF to advertise the default route (0.0.0.0/0) to its neighbors, regardless if it is present in the routing table or not
RT1(config-router)# passive-interface interface or RT1(config-router)# passive-interface default RT1(config-router)# no passive-interface interface	Tells OSPF not to send hello packets on certain interfaces Configure all interfaces, except the specified ones not to send hello packets
RT1(config-router)# router-id ip-address	Sets a fixed router ID

IPv4 ACL Commands

Configuring Standard IPv4 ACLs

RT1(config)# access-list access-list-number permit deny {source [source-wildcard] host {address name} any [log]}	Configures standard numbered IPv4 ACLs
RT1(config)# ip access-list standard access-list-name RT1(config-std-nacl)# [sequence-number] permit deny source [source-wildcard] host { address name} any	Configures standard named IPv4 ACLs

Configuring Extended IPv4 ACLs

RT1(config)# access-list access-list-number permit deny protocol source_matching_criteria destination_matching_criteria	Configures extended numbered IPv4 ACLs
RT1(config)# ip access-list extended access-list-name RT1(config-std-nacl)# [sequence-number] permit deny protocol source_matching_criteria destination_matching_criteria	Configures extended named IPv4 ACLs Protocol: A keyword indicating a protocol suite, such as ip, icmp, tcp, or udp. Keyword ip matches all protocols. source_matching_criteria: Matching criteria for the source IPv4 address and optionally port destination_matching_criteria: Matching criteria for the destination IPv4 address and optionally port Specify a single port using syntax operand eq (equal), lt (less than), gt (greater than), or neq (not equal), followed by port number (for instance 80), or the protocol name (for well-known protocols, such as www)
RT1(config-if)# ip-access-group {access-list-number access-list-name} {in out}	Links the ACL to an interface



NAT Commands

Configuring Static Translation of Inside Source Addresses

RT1(config)#ip nat inside source static local-ip global-ip	Establishes static translation between an inside local address and an inside global address
---	---

Configuring Dynamic Translation of Inside Source Addresses

RT1(config)#ip nat pool name start-ip end-ip {netmask netmask prefix-length prefix-length }	Defines a pool of global addresses to be allocated as needed
RT1(config)#access-list access-list-number permit source [source-wildcard]	Defines a standard access list permitting those addresses that are to be translated
RT1(config)#ip nat inside source list access-list-number pool name	Establishes dynamic source translation, specifying the access list

Configuring IPv4 Inside PAT

RT1(config)#access-list access-list-number permit source [source-wildcard]	Defines a standard access list permitting those addresses that are to be translated
RT1(config)#ip nat inside source list access-list-number {pool name interface exit-interface} overload	Establishes dynamic source translation with overloading, specifying the access list

Connects the interface to the inside or outside network

RT1(config-if)#ip nat inside outside	Connects the interface to the inside network, which is subject to NAT or to the outside network
---	---

WLAN COMMANDS

Switch VLAN Configuration to Support WLANs

The following types of VLANs are required with WLANs:

- **Management VLAN**
 - The management VLAN is for the WLC management interface configured on the WLC
- **AP VLAN**
 - The APs that register to the WLC can use the same VLAN as the WLC management VLAN, or they can use a separate VLAN
- **Data VLAN**

Switch Port Connected to Autonomous AP Configuration

```
SW1(config-if)#description name-string
SW1(config-if)#switchport mode trunk
SW1(config-if)#switchport trunk native vlan vlan_number
SW1(config-if)#switchport trunk allowed vlan vlan_list
```

System Monitor COMMANDS

Syslog messages format

RT1(config)#service sequence-numbers	Sequence numbers are not enabled by default Example: 000047: Apr 10 05:24:07.660: %SYS-5-CONFIG_I: Configured from console by console
RT1(config)#no service timestamps	Time stamps are enabled by default because it is much easier to identify the problem in a chronological order if you can see the time stamps on syslog messages Example: 000048: %SYS-5-CONFIG_I:



Configured from console by console	
Syslog messages format	
RT1(config)# logging {hostname ip-address}	Identifies a syslog server host to receive logging messages
RT1(config)# logging host {hostname ip-address}	Accomplishes the same thing as the logging IP-address command, except it allows you to change the default port and protocol
RT1(config)# logging trap severity	Limits the syslog messages that are sent to the syslog server. It limits the messages based on severity
Severity Level Explanation Emergency (severity 0) System is unusable Alert (severity 1) Immediate action needed Critical (severity 2) Critical condition Error (severity 3) Error condition Warning (severity 4) Warning condition Notification (severity 5) Normal but significant condition Informational (severity 6) Informational message Debugging (severity 7) Debugging message	
RT1(config)# logging source-interface interface	Identifies which interface is used as source IP address, when syslog messages will be sent

TCP/IP Applications COMMANDS

HTTP service

RT1(config)# no ip http server	Disable the HTTP service on a router
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Troubleshooting Tools

Internet Control Message Protocol

DEVICE# ping ip-address domain name	A successful ping to an IPv4 address means that the endpoints have basic IPv4 connectivity between them
DEVICE# ping ip ip-address source type interface-id	Extended Ping using specific gateway interface or source IP address
DEVICE# traceroute ip-address domain name	The results of traceroute to an IPv4 address can help you determine how far along the path data can successfully reach

Verifying the integrity of the downloaded IOS image

DEVICE# verify /md5 flash:/{file} https://www.cisco.com/c/en/us/support/docs/ios-nx-os-software/ios-software-releases-110/15171-ntpassoc.html	Computes the MD5 digests of the loaded IOS image. Verify that the digest matches that which is posted on Cisco site
--	---

Connecting a client with Telnet and SSH

DEVICE# telnet ip-address hostname	Log on to a host that supports Telnet
DEVICE# ssh ip-address	Starts an encrypted session with a remote networking device



DHCP for IPv4 COMMANDS

Configuring a Router as an IPv4 DHCP Server

RT1(config)# ip dhcp pool name	Enter the DHCP pool configuration mode
RT1(dhcp-config)# network network-number [mask / prefix-length]	Defines addresses in the DHCP pool
RT1(dhcp-config)# default-router address	Specifies the IP address of the default router for a DHCP client
RT1(dhcp-config)# dns-server address	Specifies the IP address of a DNS server
RT1(dhcp-config)# domain-name domain	Specifies the domain name for the DHCP client
RT1(dhcp-config)# lease {days [hours] [minutes] / infinite}	Specifies the duration of the lease. The default is a one-day lease.
RT1(config)# ip dhcp excluded-address ip-address [last-ip-address]	Exclude the range of IPv4 addresses from the DHCP assignment
RT1(dhcp-config)# import all	Import Dynamic Host Configuration Protocol (DHCP) option parameters into the DHCP server database
RT1(dhcp-config)# client-identifier unique-identifier	MAC-address format: 01aa.aabb.bbccc.cc Add 01 before the MAC Address as it is connected using Ethernet media and the remaining bytes represent the MAC address of the DHCP client
RT1(dhcp-config)# client-name name	Specifies the name of the client using any standard ASCII character. For BOOTP only.

Configuring an IPv4 DHCP Relay agent

RT1(config-if)# ip helper-address address	Issued on the interface where the DHCP broadcasts are received
--	--

DHCP for IPv6 COMMANDS

Configuring SLAAC on a router

RT1(config)# ipv6 unicast routing	Enter the DHCP pool configuration mode
--	--

Configuring IPv6 Stateless DHCPv6 on a router

RT1(config)# ipv6 unicast-routing	Enables IPv6 routing
RT1(config)# ipv6 dhcp pool poolname	Configures a Dynamic Host Configuration Protocol for IPv6 (DHCPv6) configuration information pool and enters DHCPv6 pool configuration mode
RT1(dhcp-config)# dns-server ipv6-address	Specifies the Domain Name System (DNS) IPv6 servers available to a DHCPv6 client
RT1(dhcp-config)# domain-name domain	Configures a domain name for a DHCPv6 client
RT1(config-if)# ipv6 dhcp server poolname [rapid-commit] [preference value] [allow-hint]	Enables DHCPv6 on an interface
RT1(config-if)# ipv6 nd other-config flag	Sets the "other stateful configuration" flag in IPv6 router advertisements (RAs)

Configuring IPv6 Stateful DHCPv6 on a router

RT1(config)# ipv6 unicast-routing	Enables IPv6 routing
RT1(config)# ipv6 dhcp pool poolname	Configures a Dynamic Host Configuration Protocol for IPv6 (DHCPv6) configuration information pool and enters DHCPv6 pool configuration mode
RT1(dhcp-config)# address prefix <specify address prefix xxxx:xxxx:xxxx:xxxx:./64>	Indicates the pool of addresses to be allocated by the server
RT1(dhcp-config)# dns-server ipv6-address	Specifies the Domain Name System (DNS) IPv6 servers available to a DHCPv6 client
RT1(dhcp-config)# domain-name domain	Configures a domain name for a DHCPv6 client



	client
RT1(config-if)# ipv6 dhcp server poolname [rapid-commit] [preference value] [allow-hint]	Enables DHCPv6 on an interface
RT1(config-if)# ipv6 nd managed-config-flag	sets the "managed address configuration flag" in IPv6 router advertisements
RT1(config-if)# ipv6 nd prefix default no-autoconfig	The A flag is manually changed from 1 to 0 (disabling SLAAC) The A flag can be left at 1, but some client operating systems such as Windows will create a GUA using SLAAC and get a GUA from the stateful DHCPv6 server. Setting the A flag to 0 tells the client not to use SLAAC to create a GUA.
Configuring an IPv6 DHCP Relay agent on a router-interface	
RT1(config-if)# ip helper-address address	Issued on the interface where the DHCP broadcasts are received

Cisco Discovery Protocol (CDP) COMMANDS

Disable or enable CDP Globally

RT1(config)# [no] cdp run	Enables or disables CDP Globally
----------------------------------	----------------------------------

Disable or enable CDP on interface only

RT1(config)# interface type interface-id	Enables or disables Cisco Discovery Protocol on an interface
RT1(config-if)# [no] cdp enable	

LLDP (Link Layer Discovery Protocol) COMMANDS

Disable LLDP Globally

RT1(config)# [no] lldp run	Enables or disables LLDP globally
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Disable or enable CDP on interface only

RT1(config)# interface type interface-id	Enables or disables LLDP on an interface
RT1(config-if)# [no] lldp transmit	
RT1(config-if)# [no] lldp receive	

Static Route Configuration COMMANDS

Host Static Routes

RT1(config)# ip route prefix [mask prefix-length] ip-address [distance]	• prefix: Specifies the IP route prefix for the destination. • mask: Specifies the prefix mask for the destination • ip-address: Specifies the IP-address of the next hop that can be used to reach that network
--	--

Configuring IPv4 Static Routes

RT1(config)# ip route prefix [mask prefix-length] [ip-address interface] [distance]	Configures a static route and the interface for this static route. You can optionally configure the next-hop address. The preference value sets the administrative distance. The range is from 1 to 255. The default is 1
Example: ip route 131.108.0.0 255.255.0.0 131.108.6.6	
RT1(config)# ip route 0.0.0.0 0.0.0.0 [outgoing-interface] next-hop	0.0.0.0: Matches all destination addresses that do not match any other route in the routing table 0.0.0.0: Matches all subnet masks

Configuring IPv6 Static Routes

RT1(config)# ipv6 route ipv6-network/ipv6-mask [outgoing-interface] ipv6-next-hop	Configures an IPv6 static route
RT1(config)# ipv6 route ::/0 [outgoing-interface] ipv6-next-hop	Configures a default IPv6 route. When the next hop is a link-local address, the outgoing interface must be specified.



STP COMMANDS

Spanning Tree Priority Root Bridge Optimization

SW1(config)# spanning-tree vlan <i>vlan_number</i> root {primary secondary}	Configure the switch as the root bridge or secondary per VLAN
--	---

Rapid Spanning Tree Protocol

SW1(config)# spanning-tree mode rapid-pvst	Enables rapid spanning-tree
SW1(config)# spanning-tree portfast default	Enables portfast on all non-trunking ports
Warning: Never enable PortFast on ports connecting to other switches or hubs. This could lead to bridging loops and broadcast storms since the switch port never goes through the listening and learning state of STP	
SW1(config-if)# spanning-tree portfast	Enables portfast per-interface

BPDu guard

SW1(config)# spanning-tree portfast bpduguard default	Enables bpduguard on all ports
Warning: BPDu guard should never be implemented on Uplink ports where Root Bridge is located. In case of multiple uplinks, BPDu guard must be disabled to accept legitimate BPDUs.	
SW1(config-if)# spanning-tree bpduguard enable	Enables bpduguard per-interface

Port Security COMMANDS

Configuring Port Security

SW1(config-if)# switchport port-security	Enables port security on an interface
SW1(config-if)# switchport port-security maximum <i>value</i>	Sets the maximum number of secure MAC addresses
SW1(config-if)# switchport port-security mac-address <i>mac-address</i>	Allows only the device with defined MAC address to connect
SW1(config-if)# switchport port-security mac-address sticky	MAC addresses are learned dynamically
SW1(config-if)# switchport port-security violation {protect restrict shutdown }	Sets the violation mode. This table summarizes the effect of each violation mode. The default mode is shutdown
SW1(config-if)# switchport port-security aging type {absolute inactivity}	Sets the aging type
Absolute: The secure addresses on the port are deleted after the specified aging time. Absolute aging is the default type if aging is enabled	
Inactivity: The secure addresses on the port are deleted only if the secure addresses are inactive for a specified aging time. Aging time is specified in minutes	
SW1(config-if)# switchport port-security aging time	The maximum aging time before the secure addresses will "expire" is 24 hours - 1440 minutes

Mitigating Attacks COMMANDS

Mitigating VLAN Attacks

SW1(config-if)# switchport mode trunk	
SW1(config-if)# switchport nonegotiate	

Mitigating DHCP-related attacks

SW1(config)# ip dhcp snooping	Enables DHCP snooping globally
SW1(config)# ip dhcp snooping vlan <i>vlan-list</i>	Enables DHCP snooping on selected VLANs
SW1(config)# no ip dhcp snooping information option	Will disable the switch from inserting



	option 82 in the DHCP packets. This will offer better compatibility with DHCP servers because some servers drop packets with option 82
SW1(config-if)#ip dhcp snooping trust	Configures trusted interfaces, because untrusted is the default configuration
Mitigate the chances of ARP spoofing	
SW1(config)#ip arp inspection vlan-id	Enables ARP inspection on selected VLANs
SW1(config-if)#ip arp inspection trust	Enables DAI on an interface and set the interface as a trusted interface
Mitigating STP Attacks	
SW1(config-if)#spanning-tree guard root	Prevents a switch from becoming a root bridge on configured ports
Mitigating Broadcast storms	
SW1(config-if)#storm-control {broadcast multicast unicast} level {level [level-low] bps bps [bps-low] pps pps [pps-low]}	Sets a limit to the number of broadcast, multicast or unknown unicast traffic and an action when the threshold is exceeded. Default action will drop exceeding traffic.
SW1(config-if)#storm-control action {shutdown trap}	Specifies the action to be taken when a storm is detected. The default is to filter out the traffic and not to send traps
Shutdown: error-disable the port during a storm	
Trap: generates an SNMP trap when a storm is detected	

Network Time Protocol COMMANDS

Configure Software Clock

RT1#clock set hh:mm:ss { day month month day } year	Changes the software clock settings
RT1(config)#clock timezone zone-name hours-offset [minutes-offset]	Sets the time zone for display Zone-name = CET (Central Europe Time, as UTC plus 1 hour) Hours offset from Coordinated Universal Time (UTC). Range is from -23 to +23
RT1(config)#clock summertime zone { date { date month year hh:mm date month year hh:mm month date year h h:mm month date year hh:mm } recurring week day month hh:mm week day month hh:mm } [offset]	configure the system to switch automatically to summer time (daylight saving time) recurring: indicates that summer time should start and end on the corresponding specified days every year
RT1#clock update-calendar	Updates the hardware clock with a new software clock setting. The hardware system clock runs continuously, even if the router is powered off or rebooted
RT1#clock read-calendar	Copies hardware clock settings to the software clock. The hardware system clock is separate from the software clock settings, which are erased when the router is power cycled or rebooted

Configuring NTP

RT1(config)#ntp master [stratum]	Configures a Cisco IOS device as an NTP master clock to which peers synchronize themselves
RT1(config)#ntp server ip-address	Configures a Cisco device to use the specified NTP server



Loading and Managing System Images Files COMMANDS

Locating Cisco IOS Image Files

RT1# show file systems	Lists all the available file systems on a Cisco Integrated Services Router
RT1(config)# boot system flash: {file}	A global configuration command that allows you to specify the source for the Cisco IOS Software image to load

Saving the configuration to a TFTP-server

DEVICE# copy running-config tftp:	Copies the running configuration to a TFTP server to create a Cisco IOS Image backup
--	--



SHOW COMMANDS

Verify Basic Configuration

SW1#show version	Shows information about the device and its interfaces, RAM, NVRAM, flash, IOS, etc.
SW1#show running-config begin exclude include partition	Shows the current configuration file stored in RAM.
SW1#show startup-config	Shows display the saved configuration stored in config.text located in NVRAM
SW1#show running-config interface type interface-id VLAN number	You can specify an interface directly to the show running-config command
SW1#show banner-motd	Displays the system-defined banner

Inspect Privilege Level

RT1#show privilege	Displays the current privilege level
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Inspect TCP/IP Applications

RT1#show control-plane host open-ports	Verify the services that are running
RT1#show users	Displays the remote logins

Verify configured DHCP address pools

RT1#show ip dhcp pool	Verify information about the configured DHCP address pools
RT1#show ip dhcp binding	Displays a list of all IPv4 address-to-MAC bindings

Verify configured DHCPv6 address pools

RT1# show ipv6 interface [brief] [type number] [prefix]	Display's the usability status of interfaces configured for IPv6
RT1#show ipv6 dhcp pool	Display's the Dynamic Host Configuration Protocol (DHCP) for IPv6 configuration pool information
RT1#show ipv6 dhcp binding	Display's automatic client bindings from the Dynamic Host Configuration Protocol (DHCP) for IPv6 server binding table

Checking Interface Configuration and Status

SW1#show ip[v6] interface [brief]	Shows an overview of all interfaces, their physical status, protocol status and IP-address
SW1#show interfaces	Details about all the interfaces
SW1#show interfaces type interface-id	Details for a specific interface
SW1#show interface status	Displays the interface line status
RT1#show protocols type interface-id	Brief details about a particular interface
RT1#show ip protocols	Display the status of the configured dynamic routing protocols on a router

View Cisco Discovery Protocol (CDP) information

RT1#show cdp entry	Information for specific neighbor entry
RT1#show cdp interface [detail]	CDP interface status and configuration
RT1#show cdp neighbors	CDP neighbor entries
RT1#show cdp traffic	CDP statistics

Verify LLDP (Link Layer Discovery Protocol)

RT1#show lldp neighbors	Displays information about neighbors, including device ID, interface type and number, hold time settings, capabilities, and port ID
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View the ARP table

RT1#show ip arp [ip-address] [host-name] [mac-address] [interface type number]	Display the ARP table on a Cisco IOS router
RT1#show arp	Display the ARP table on a Cisco IOS router



View Logging	
RT1#show logging	Verify logging settings on networking devices
View neighbor discovery table	
RT1#show ipv6 neighbors	Includes the IPv6 address of the neighbor, age in minutes, the MAC address, the state, and the interface through which the [neighbor is reachable
Verifying IPv6 Static Route Configuration	
RT1#show ipv6 route static	Verify the IPv6 static route configuration in the routing table
RT1#show ipv6 static [ipv6-address / ipv6-prefix/prefix-length][interface interface-type interface-number] [recursive] [detail] or show ipv6 route static	Verify the static IPv6 route
Verifying VLANs	
SW1#show vlan	Displays information on all configured VLANs
SW1#show vlan id vlan_number	Displays information about a particular VLAN
SW1#show vlan name vlan-name	Displays information about a particular VLAN
SW1#show vlan brief	Displays one line for each VLAN with the VLAN name, status, and its ports
Verifying the VLAN configuration of an interface	
SW1#show interfaces interface-id switchport	Displays information on all configured VLANs
Verifying VTP configuration	
SW1#show vtp status	Verify VTP operation and status
Verifying MAC addresses	
SW1#show mac address-table	Verify which MAC addresses belong to which port and VLAN
SW1#show mac address-table vlan vlan-id	To see the MAC addresses that have been learned on a particular VLAN
Verifying IPv4 routing table	
RT1#show ip route	Verify the IPv4 routing table
Verify Single-Area OSPF	
RT1#show ip ospf interface brief	Verifies the interfaces that are enabled for OSPF
RT1#show ip ospf interface interface-id	Verifies all OSPF-related configuration on an interface
RT1#show ip ospf neighbor	Displays the OSPF neighbor information on a per-interface basis
Verify EtherChannel Configuration	
SW1#show interface port-channel number	Displays the general status of the logical port channel interface that represents the aggregated link
SW1#show etherchannel summary	Displays one line of information per port channel and is particularly useful when several port channel interfaces are configured on the same device
SW1#show etherchannel number port-channel	Displays information about the specific port channel interface
Verify STP information	
SW1#show spanning-tree	Displays STP information
SW1#show spanning-tree vlan vlan-id	Displays STP information for a specific VLAN
Verify HSRP information	
SW1#show standby [type number [group]] [all brief]	Displays Hot Standby Router Protocol (HSRP) information
Verify the configured access-list	
RT1#show access-lists access-list-number	Displays the configured access-list



```
RT1#show running-config | include  
access-list {access-list-number}
```

Filters out only numbered access lists from the show running-config



Verify Network Address Translation	
RT1#show ip nat statistics	Shows the specific NAT Statistics and translation that have occurred for the configured NAT mapping
RT1#show ip nat translations	Displays the number of NAT translation is shown
RT1#clear ip nat translation	Clears NAT entries; you can specify a single entry by the global and local address or by TCP and UDP translations (including ports), or you can use an asterisk (*) to clear the entire table
Display the state of logging (syslog)	
RT1#show logging	Displays the state of syslog error and event logging, including host addresses, and whether console logging is enabled. This command also displays Simple Network Management Protocol (SNMP) configuration parameters and protocol activity
Verifying NTP Status	
RT1#show ntp associations https://www.cisco.com/c/en/us/support/docs/ios-nx-os-software/ios-software-releases-110/15171-ntpassoc.html	Determines if NTP is working and communicating properly
RT1#show ntp status	Checks the status of the NTP server or peer
Verifying running Services	
RT1#show control-plane host open-ports	Displays the list of UDP or TCP ports that the device is listening on and to determine which services need to be disabled
or	
RT1#show ip ports all	IOS versions prior to IOS-XE use the show control-plane host open-ports command
Verifying Port Security	
SW1#show port-security	Displays port security settings for the switch, including violation count, configured interfaces, and security violation actions
SW1#show port-security interface interface-id	Displays port security settings for the specified interface, including the maximum allowed number of secure MAC addresses for each interface, the number of secure MAC addresses on the interface, the number of security violations that have occurred, and the violation mode
SW1#show port-security [interface interface-id] address	Displays all the secure MAC addresses that are configured on all switch interfaces, or on a specified interface, with aging information for each address
Verifying DHCP Snooping	
SW1#show ip dhcp snooping	Shows only trusted ports or the ports with a configured rate limit
SW1#show ip dhcp snooping binding	Displays all the known DHCP bindings that have been learned on a switch
Verifying Dynamic ARP Inspection	
SW1#show ip arp inspection	Displays the status of the DAI configuration
SW1#show ip arp inspection interfaces	Displays the status of the DAI configuration
SW1#show ip arp inspection log	Reviews DAI activity
SW1#show ip arp inspection statistics	Reviews DAI activity



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